Traffic Volume Prediction Using Machine Learning

MINOR PROJECT - 1

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Year : 3rd (5th Sem)

Team : 46

Introduction:-

Traffic congestion is one of the most critical challenges in modern cities. With increasing numbers of vehicles on the road, predicting traffic flow has become essential for improving road safety, reducing travel time, enhancing traffic management, and planning future infrastructure. Traditional prediction methods often fail because traffic depends on multiple variables such as time, weather, and human behavior.

With advancements in artificial intelligence, machine learning models are capable of identifying hidden patterns within large datasets. This project uses a real-world dataset and machine learning techniques to predict traffic volume more accurately and efficiently.

Problem Statement :-

Urban road networks experience fluctuating traffic throughout the day. Factors like working hours, holidays, weather conditions, and seasonal changes contribute to unpredictable traffic flow. Manual forecasting methods cannot analyze complex relationships among these variables, leading to inaccurate predictions.

The core problem addressed in this project is:

"Can we accurately predict the number of vehicles passing through a road based on time and environmental conditions?"

By solving this, we aim to assist transportation planning authorities, navigation systems, and daily commuters with better predictions.

Economic Drain

Billions lost annually in productivity and fuel.

Environmental Impact

Increased emissions and air pollution.

Quality of Life

Longer commutes, higher stress levels for citizens.

Proposed Solution : -

Our solution is to design an **end-to-end machine learning system** that predicts hourly traffic volume using historical traffic data, weather information, and time-based features. After preprocessing the dataset and applying feature engineering, we train three ML models and select the best performer (Random Forest).

The final model is deployed through a Flask API, and an interactive web application enables users to input conditions and instantly obtain traffic predictions.

Project Objectives: Enhancing Predictive Capabilities

The major objectives of this project include:

Dataset Acquisition & Exploration:
Collect and deeply analyze the Metro Interstate Traffic Volume dataset to understand its structure and patterns.

Data Preprocessing & Cleaning:
Handle missing values, ensure consistency in weather and time fields, extract useful features from date-time, and prepare high-quality input data for modeling.

Model Development:
Build three machine learning models — Linear Regression, XG Boost and Random Forest — to evaluate their performance on traffic volume prediction.

Model Comparison:
Compare models using metrics such as MAE, RMSE, and R^2 to identify the model that provides the highest accuracy.

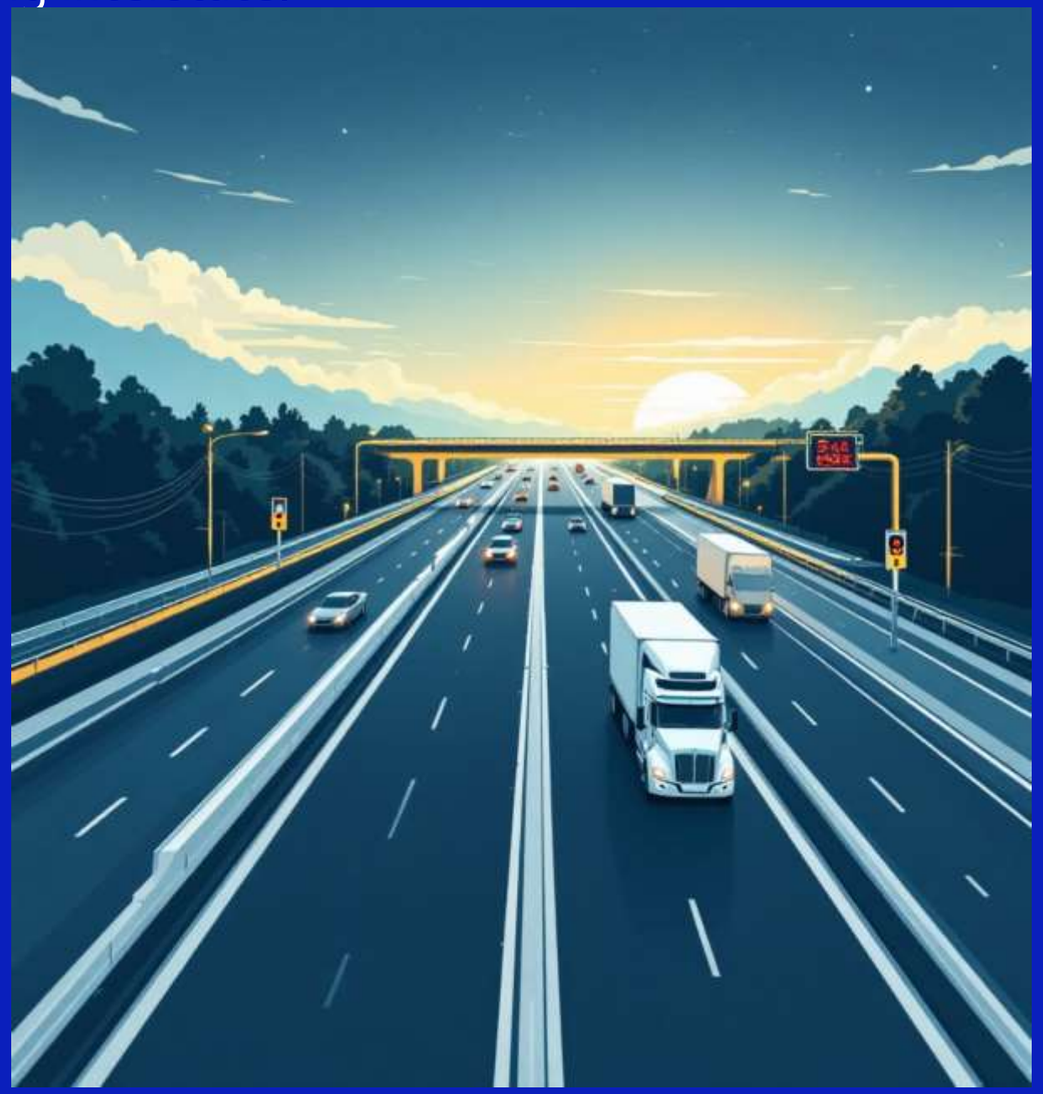
Deployment & Web Interface:
Create a clean and interactive **Streamlit web application** where users can input time and weather details and instantly receive the predicted traffic volume along with traffic condition categories.

Dataset: Metro Interstate Traffic Volume:-

The dataset used in this project is sourced from Kaggle: **Metro Interstate Traffic Volume**. It contains traffic data collected from Automated Traffic Recorder (ATR) stations along Interstate.

Dataset Highlights :

- Total Records: Approximately **48,000 rows**
- Columns include:
  - **date_time:** Combined date and time of observation.
  - **temp:** Temperature in Kelvin.
  - **rain_1h & snow_1h:** Precipitation levels.
  - **clouds_all:** Cloud cover percentage.
  - **weather_main / weather_description:** Weather conditions.
  - **traffic_volume:** Number of vehicles per hour.



Tools & Technologies :-

Our project utilizes a robust set of tools and technologies, ensuring efficient data processing, model development, and seamless application deployment.

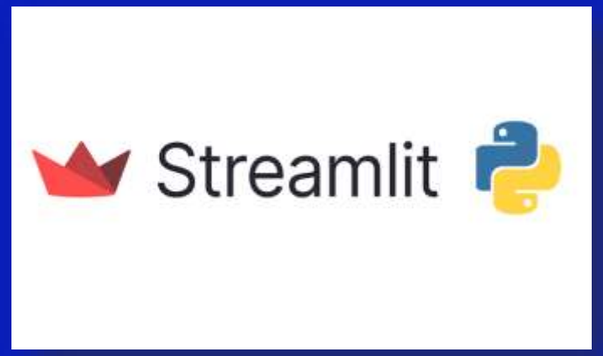
Python
Primary language for data manipulation and ML.

Scikit-learn
Linear regression, random forest

Jupyter Notebook
Interactive environment for analysis and prototyping.

Flask
Web framework for API development

Matplotlib/Seaborn
For data visualization.

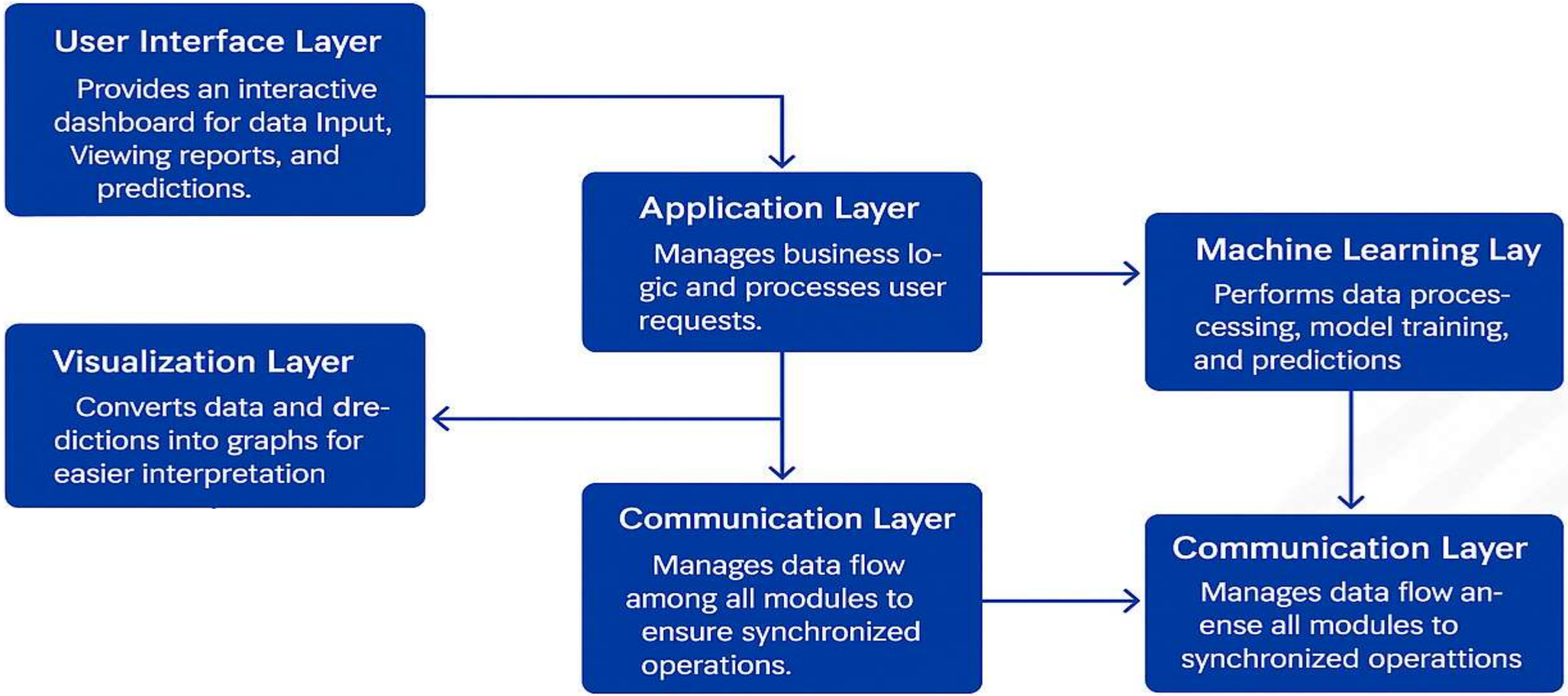


Streamlit
For Dashboard

Pandas
Data Manipulation

Numpy
For Numerical Operation

SYSTEM ARCHITECTURE



Data Cleaning Steps: Preparing for Precision:-

Raw datasets often contain inconsistencies. Hence, preprocessing is essential before training models.

- **Datetime Conversion :** Converted date time to a proper datetime object and extracted useful features such as hour, weekday, and month.

- **Handling Missing Values** : Checked for NaNs in weather and traffic columns and handled them appropriately.

- **Feature Standardization** : Ensured numeric fields like temperature and precipitation were within expected ranges.

- **Feature Creation :** Added a new column: is weekend, to distinguish between weekday and weekend behavior.

- **Outlier Detection :** Identified and analyzed unusual spikes or drops in traffic volume to maintain prediction accuracy.

EXPLORATORY DATA ANALYSIS :-

Visual analysis provided several key insights:

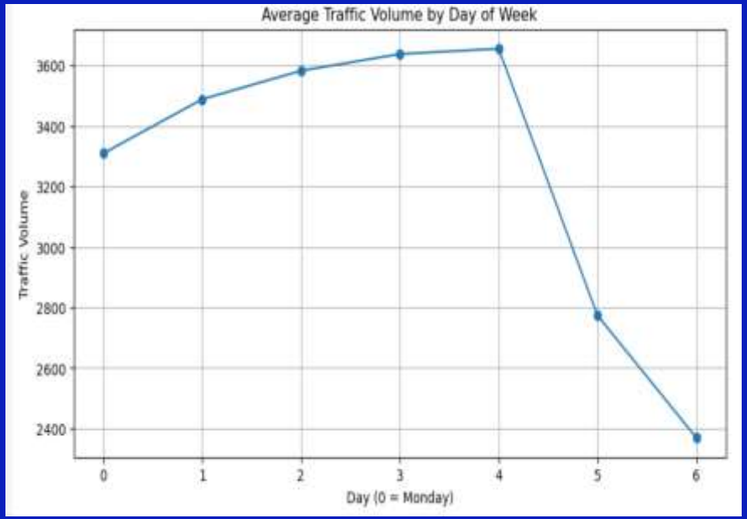
Hourly Traffic Pattern: Clear morning and evening rush hours were visible. Traffic drops significantly late at night.

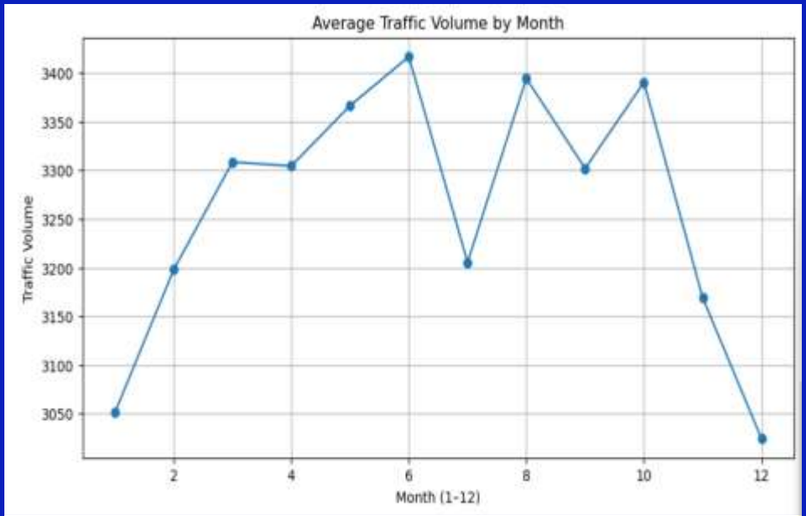
Weekday Trends: Weekdays showed heavier traffic compared to weekends, confirming office-hour patterns.

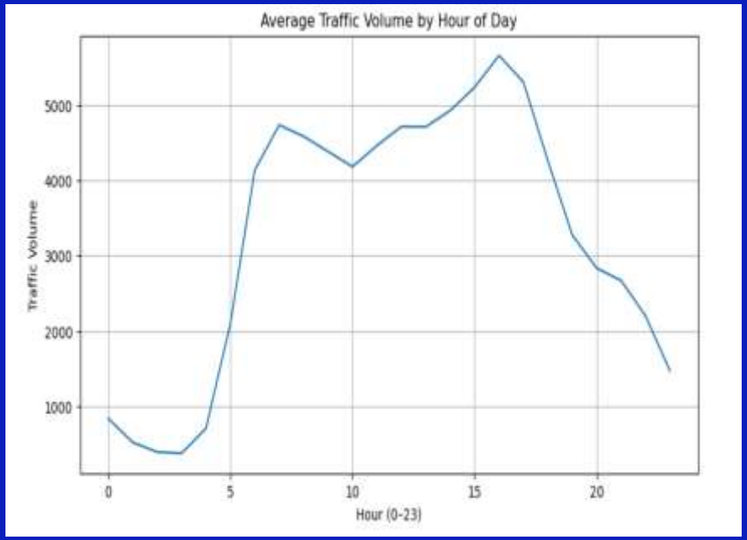
Weather Impact:
Heavy rain, snowfall, and overcast conditions tended to reduce traffic volume.

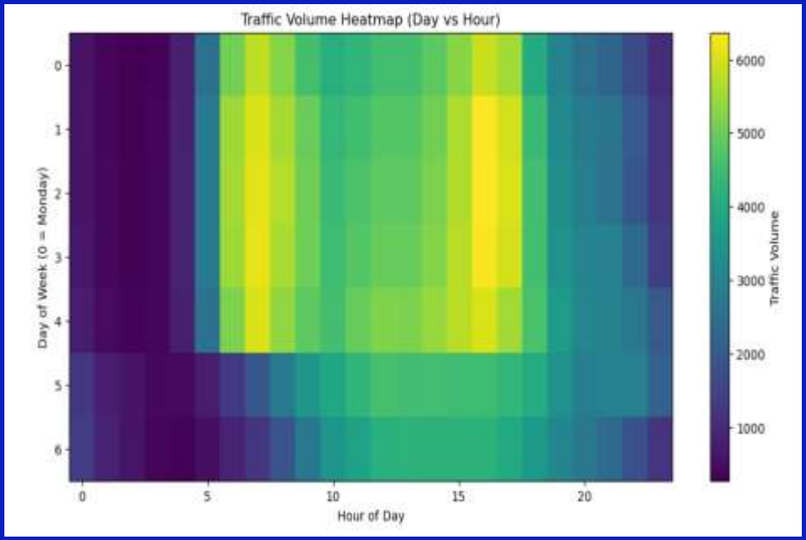
Monthly Behavior:
Certain months had distinct seasonal traffic patterns influenced by weather.

Heatmap Analysis:
A correlation heatmap revealed how features like temperature, hour, and weekday correlated with traffic volume.
These insights helped shape the feature engineering and model training steps.

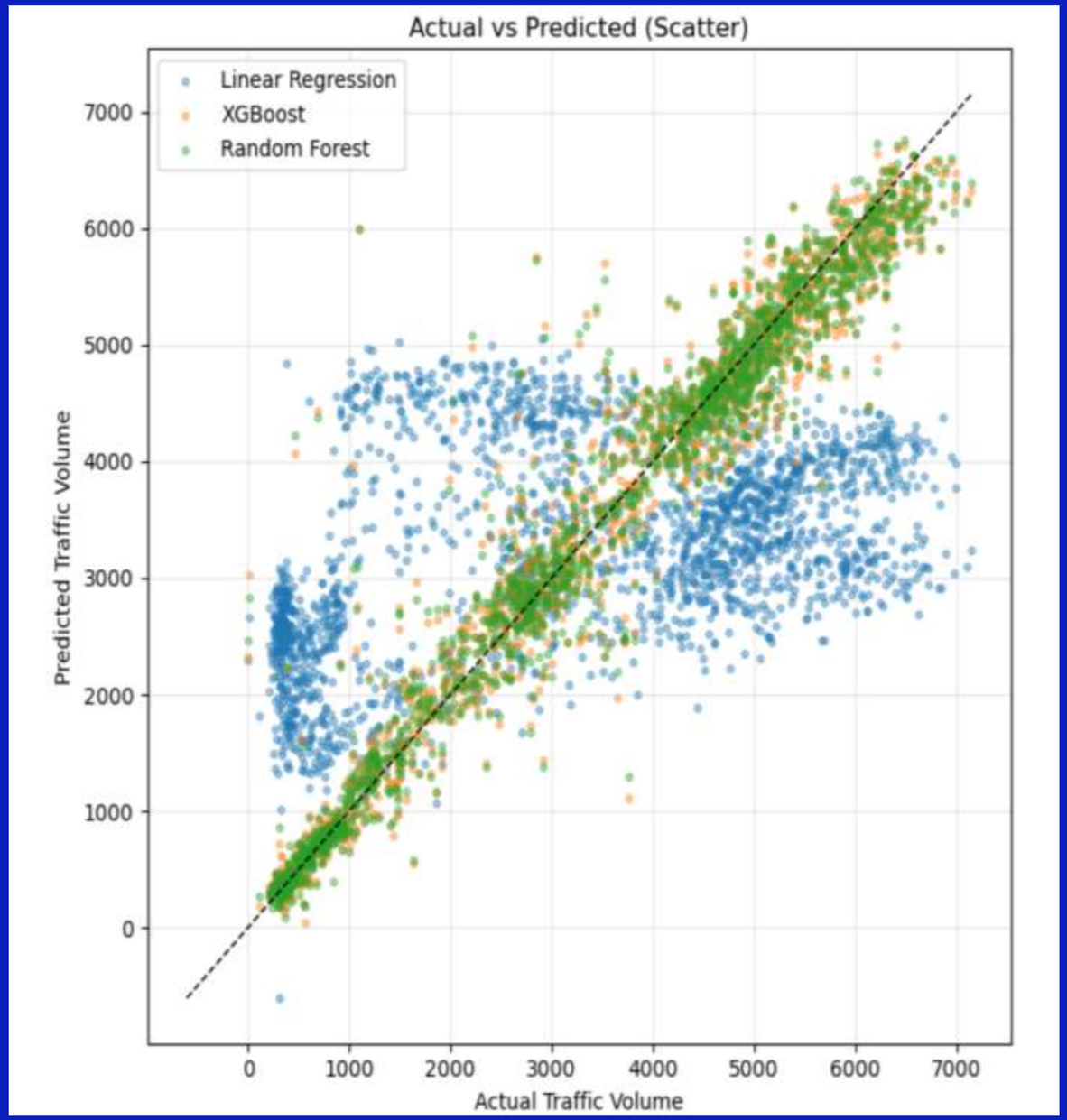








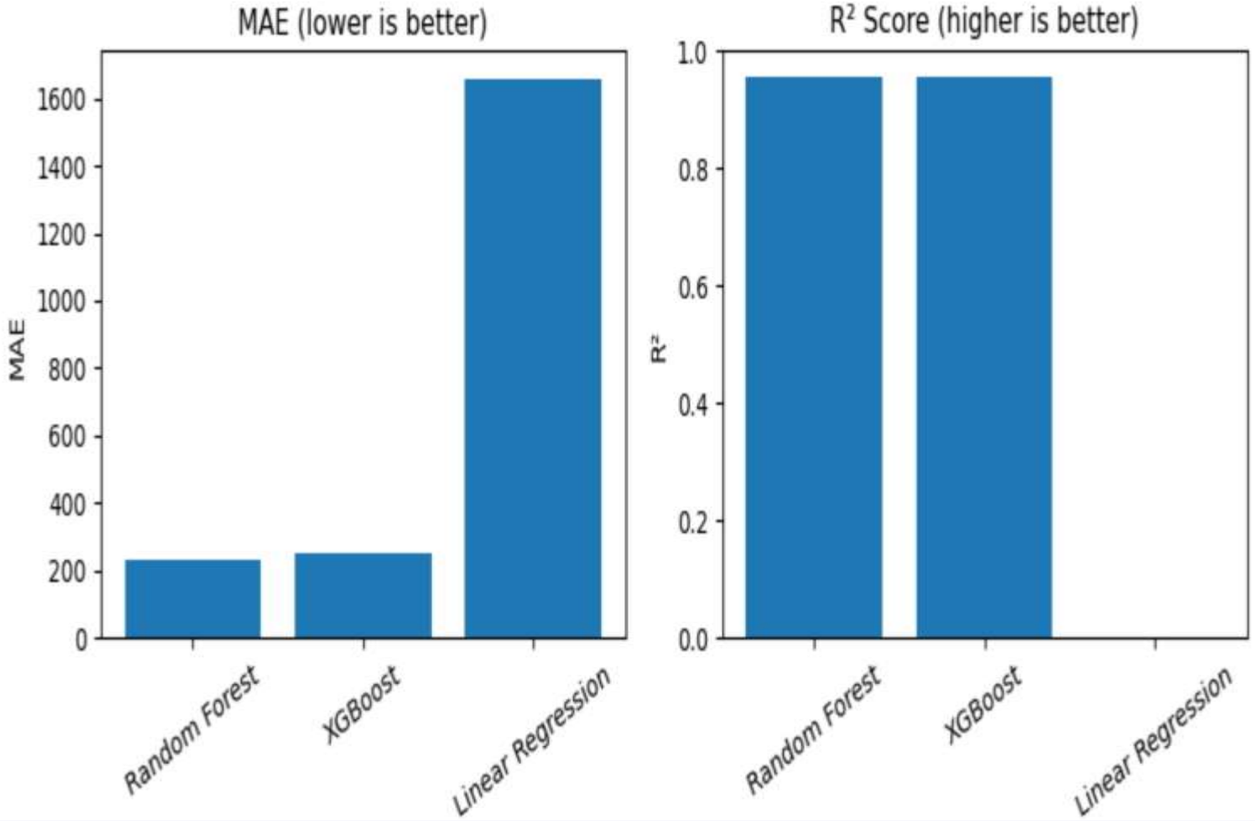
OUTPUT :



Model comparison (sorted by MAE):

	Model	MAE	R2
0	Random Forest	229.599504	0.956237
1	XGBoost	249.052597	0.955497
2	Linear Regression	1656.280705	-9.208578

Saved comparison to model_comparison.csv



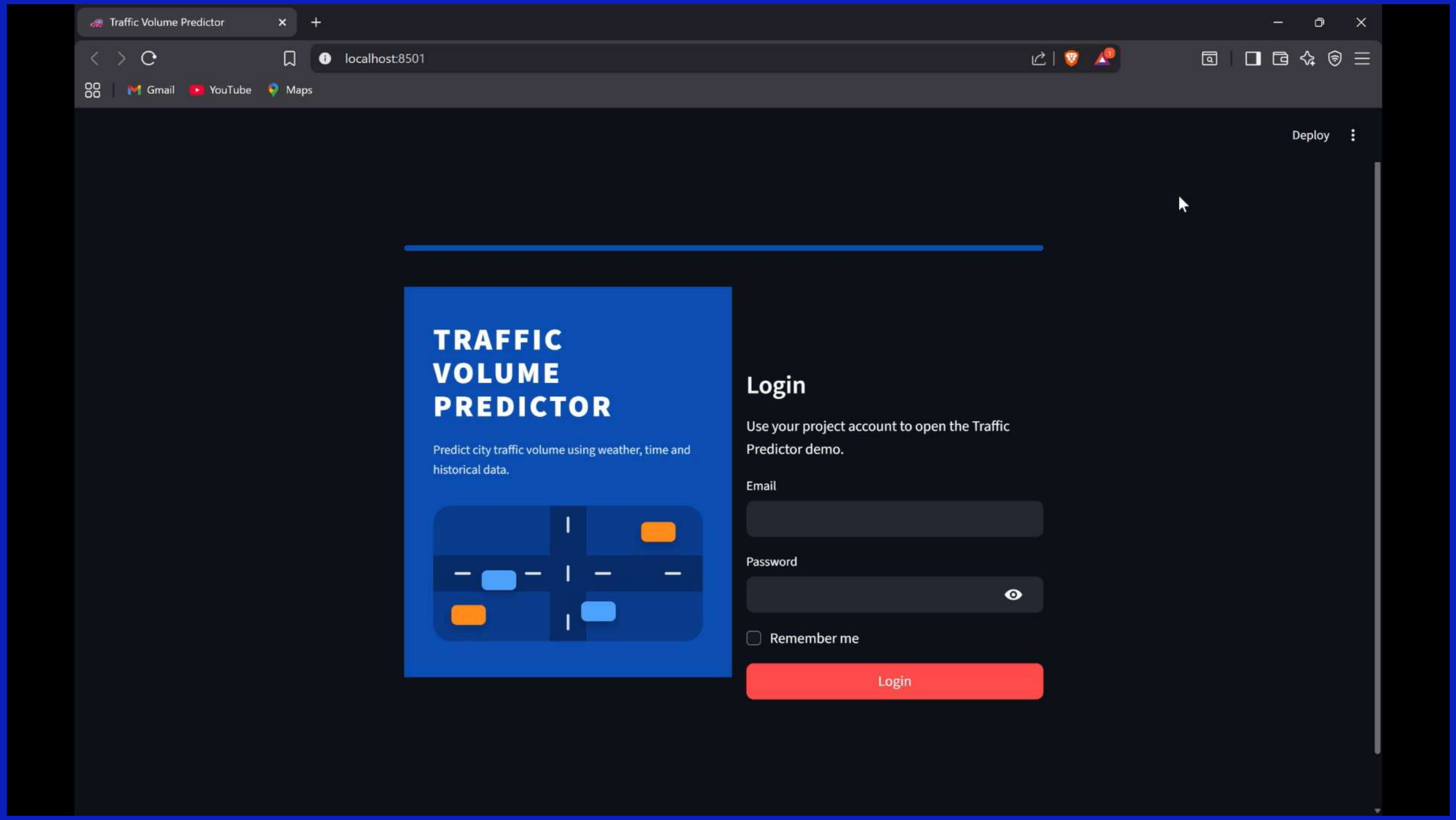
MODEL COMPARISON :

Models were compared using three key metrics:

- **MAE (Mean Absolute Error):** Measures average absolute error.
- **RMSE (Root Mean Squared Error):** Penalizes large errors.
- **R^2 (Coefficient of Determination):** Measures how much variance is explained.

Metric	Linear Regression	XG Boost	Random Forest
MAE:	1656.28	249.05	229.59
RMSE:	2700	353	334
R^2 Score:	-9.208578	0.955497	0.956237

SYSTEM DEMONSTRATION :



Benefits & Future Enhancements:

Key Benefits :

- **Accurate Prediction:** Provides precise hourly traffic forecasts
- **Optimized Routes:** Empowers commuters to plan efficient journeys.
- **Weather-Aware Insights:** Offers predictions considering environmental factors.
- **Urban Planning:** Invaluable for city planners and logistics operations..

Future Enhancements

- **Live API Integration:** Connect with real-time traffic data sources.
- **Google Maps Integration:** Seamless route suggestions.
- **Time-Series Models:** Implement advanced forecasting techniques.
- **Notifications:** Alert users to high-traffic conditions.
- **Mobile App:** Real-time access on the go.

CONCLUSION:

1. The system provides accurate traffic predictions using machine learning.
The developed model analyzes historical traffic data along with weather and time-based factors to generate reliable hourly traffic forecasts. These predictions help users understand congestion levels before traveling, reducing uncertainty and improving planning.

2. Random Forest proves to be the most reliable and accurate model.
Among all tested models (Linear Regression, XGBoost, Random Forest), Random Forest achieved the lowest MAE and highest R^2 score. Its ability to capture non-linear relationships in traffic behavior makes it ideal for real-world traffic prediction.

3. The Streamlit dashboard offers a simple and intuitive user interface.
The dashboard allows users to easily input weather and time information and instantly view predicted traffic levels. Its clean layout, interactive elements, and user-friendly design ensure that even non-technical users can access and interpret predictions effortlessly.

4. The project is scalable and can integrate real-time data in the future.
The architecture allows easy integration of live traffic APIs, advanced forecasting techniques, or mobile interfaces. Additional modules—such as real-time alerts, route optimization, or time-series forecasting—can be implemented without changing the core system.

5. The system supports commuters and traffic authorities in better planning.
Accurate predictions help commuters avoid peak-hour congestion, saving time and fuel. Traffic authorities can use these insights for traffic flow management, road planning, and scheduling maintenance activities during low-traffic periods.

Thank You!